CHAPTER: 7
DATA DICTIONARY

7.1 Introduction

Data Dictionary is an important part of a project or system which contains all definition of elements in the system. In Data Dictionary you will find a list of all elements composing the data flowing through a system.

- The major elements of a system:
  - Data Flow
  - Data Store
  - Processes
- The data dictionary stores all details and description of these elements.
- The data dictionary provides additional information about the system.
- The data dictionary contains the data about data e.g. Student is a data and this student belongs to this course is a description of STUDENT data which is stored in data dictionary.
- Why is Data Dictionary important?
  - To manage the details in large system.
  - To communicate a common meaning for all system elements.
  - To document the features of system.
  - To determine where system changes should made.
  - To locate errors and omissions in the system.

7.2 List of Tables:

- **Table 7.2.1: user_type**
  - **Description: Stores user types of system (Admin/Organizer/Participant/Judge).**

[Table 7.2.1: user_types]

Field Name	**Data type (size)**	**Constraint**	**Description**
user_type_id	BIGINT	Primary key, Auto Increment	Unique role id
user_type	Varchar(30)	Not Null, Unique	Role Name (ADMIN/ORGANIZER/PARTICIPANT/JUDE)

- **Table 7.2.2: users**
    - **Description: Stores login and basic profile details**

[Table 7.2.2: users]

Field Name	**Data type (size)**	**Constraint**	**Description**
user_id	BIGINT	Primary key, Auto Increment	Stores the unique identifier of the user.
first_name	Varchar(100)	Not Null	User’s first name
last_name	Varchar(100)	Not Null	User’s last name
email	Varchar(150)	Not Null, Unique	User’s email
password	Varchar(255)	Not Null	Encrypted password
role	VARCHAR(30)	NOT NULL	Role label
profile_pic_url	VARCHAR(500)	NULL	Profile image url
active	BOOLEAN	NOT NULL	Account Status
otp	VARCHAR(10)	NULL	• OTP for password reset
user_type_id	BIGINT	FK -> user_types.user_type_id	Role reference

- **Table 7.2.3: user_details**
    - **Description: Stores extended user profile information**

[Table 7.2.3: users_details]

Field Name	Data type (size)	Constraint	Description
user_detail_id	BIGINT	Primary key, Auto Increment	PK, Auto Increment
user_id	BIGINT	FK -> users.user_id, UNIQUE	Owner user
bio	VARCHAR(1000)	NULL	Short profile bio
location	VARCHAR(150)	NULL	User location
linkedin_url	VARCHAR(255)	NULL	LinkedIn Profile link
role	VARCHAR(30)	NOT NULL	Role label
profile_pic_url	VARCHAR(500)	NULL	Profile image url

- **Table 7.2.4: catagories**
    - **Description: Stores hackathon categories.**

[Table 7.2.4: catagories]

Field Name	Data type (size)	Constraint	Description
category_id	BIGINT	PK, Auto Increment	Unique category id
category_nam	VARCHAR(100)	NOT NULL, UNIQUE	Category name

active	BOOLEAN	NOT NULL	Category status

- **Table 7.2.5: hackathon**
  - **Description: Stores hackathon categories.**

[Table 7.2.5: hackathon]

Field Name	Data type (size)	Constraint	Description
hackathon_id	BIGINT	PK, Auto Increment	Unique hackathon id
title	VARCHAR(200)	NOT NULL	Hackathon Title
description	TEXT	NOT NULL	Event description
status	VARCHAR(30)	NOT NULL	UPCOMING/ONGOING/COMPLETED
event_type	VARCHAR(30)	NOT NULL	ONLINE/OFFLINE/HYBRID
payment	VARCHAR(30)	NOT NULL	FREE/PAID
location	VARCHAR(200)	NULL	Event location
registration_start_date	DATE	NOT NULL	Registration start
registration_end_date	DATE	NOT NULL	Registration end
min_team_size	INT	NOT NULL	Minimum team members
max_team_size	INT	NOT NULL	Maximum team members
user_id	BIGINT	FK -> users.user_id	Organizer id
category_id	BIGINT	FK -> categories.category_id	Category id

- **Table 7.2.6: teams**
    - **Description: Stores team records for hackathons.**

[Table 7.2.6: teams]

Field Name	**Data type (size)**	**Constraint**	**Description**
team_id	BIGINT	PK, Auto Increment	Unique team id
team_name	VARCHAR(150)	NOT NULL	Team Name
created_at	DATETIME	NOT NULL	Team creation timestamp
leader_user_id	BIGINT	FK -> users.user_id	Team leader

- **Table 7.2.7: team_members**
    - **Description: Stores members belonging to teams.**

[Table 7.2.7: team_members]

Field Name	**Data type (size)**	**Constraint**	**Description**
team_member_id	BIGINT	PK, Auto Increment	Unique member row id
team_id	VARCHAR(150)	FK -> teams.team_id	Team reference

member_user_id	BIGINT	FK -> users.user_id	Member user
role	VARCHAR(30)	NOT NULL	LEADER/MEMBER

- **Table 7.2.8: hackathon_applications**
  - **Description: Stores application submissions for hackathons.**

[Table 7.2.8: team_members]

Field Name	**Data type (size)**	**Constraint**	**Description**
tapplication_id	BIGINT	PK, Auto Increment	Unique application id
team_id	BIGINT	FK -> teams.team_id	Team reference
participant_user_id	BIGINT	FK -> users.user_id	Applicant user
status	VARCHAR(30)	NOT NULL	PENDING/APPROVED/REJECTED/etc
payment_status	VARCHAR(30)	NOT NULL	WAIVED/PENDING/PAID
applied_at	DATETIME	NOT NULL	Application timestamp
submission_url	VARCHAR(500)	NULL	Project submission URL
frontend_github_link	VARCHAR(500)	NULL	Frontend repo URL
backend_github_link	VARCHAR(500)	NULL	Backend repo URL

submission_description	TEXT	NULL	Submission details

- **Table 7.2.9: judge_assignments**
    - **Description: Stores team records for hackathons.**

[Table 7.2.9: judge_assignments]

Field Name	**Data type (size)**	**Constraint**	**Description**
judge_assignment_id	BIGINT	PK, Auto Increment	Unique assignment id
hackathon_id	BIGINT	FK -> hackathon.hackathon_id	Hackathon reference
judge_user_id	BIGINT	FK -> users.user_id	Assigned judge
assigned_by_user_id	BIGINT	FK -> users.user_id	Organizer/Admin who assigned
assigned_at	DATETIME	NOT NULL	Assignment timestamp

- **Table 7.2.10: judge_scores**
    - **Description: Stores scorecards given by judges.**

[Table 7.2.10: judge_scores]

Field Name	**Data type (size)**	**Constraint**	**Description**
judge_score_id	BIGINT	PK, Auto Increment	Unique score id
application_id	BIGINT	BIGINT | FK -> hackathon_applications.application_id	Application reference
judge_id	BIGINT	FK -> users.user_id	Judge Reference
score	DECIMAL(5,2)	NOT NULL	Numeric score
remarks	VARCHAR(1000)	NULL	Judge comments